Dipole radiation appears when we can neglect $\frac{\hat{n}\cdot\vec{r}}{c}$ in argument of $\rho, \vec{J}$ (via t') and $\vec{A}$.

let T be time when charge distribution changes

let S be size of the system.

$$\Rightarrow \frac{\vec{n}\cdot\vec{r}}{c} \sim \frac{S}{c} \ll T \Rightarrow S \ll cT \sim \frac{c}{\omega} \sim \lambda$$

wavelength

$\Rightarrow$ size $\underline{\underline{S \ll \lambda}}$

$\omega \sim \frac{1}{T}$ because it is acceleration that matters!

Also, $T \sim \frac{S}{v} \Rightarrow \lambda \sim \frac{cS}{v}$

$\Rightarrow$ from $S \ll \lambda$ follows $\underline{\underline{v \ll c}}$

Thus $R_0 \gg r$ and $R_0 \gg \lambda$ (neglect $\frac{\hat{n}\cdot\vec{r}}{c}$)

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi R_0}\int d\vec{r}\, \vec{J}\left(\vec{r}, t - \frac{R_0}{c}\right) \qquad t' = t_0 - \frac{R_0}{c}$$

independent of $\vec{r}$!

but $\vec{J} = \sum_j \vec{v}_j q_j \delta(\vec{r} - \vec{r}_j(t)) = \rho\vec{v}$

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi R_0}\left(\sum_j q_j \vec{v}_j\right)_{t'} = \frac{\mu_0}{4\pi}\frac{\dot{\vec{d}}}{R_0}$$

but $\sum q_j \vec{v}_j = \frac{d}{dt}\sum_j q_j \vec{r}_j = \dot{\vec{d}}$ = dipole moment

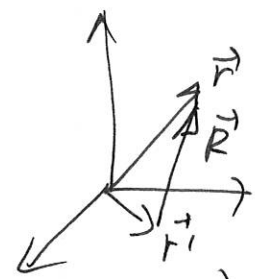
$$\vec{B} = \frac{1}{c}\dot{\vec{A}}\times\hat{n} = \frac{\mu_0}{4\pi c R_0}\left(\ddot{\vec{d}}\times\hat{n}\right)$$

$$\vec{E} = c\vec{B}\times\hat{n}$$

Quadrupole and magnetodipole radiation.

$$\vec{A} = \frac{\mu_0}{4\pi r} \int d\vec{r}\,' \; \underline{j\left(\vec{r}\,', t - \frac{|\vec{r}-\vec{r}\,'|}{c}\right)}$$

$$|\vec{r}-\vec{r}\,'| = r - \hat{n}\,\vec{r}\,'$$



Dipole radiation corresponds to complete neglect of $\hat{n}\,\vec{r}\,'$ term in $\left(t - \frac{r}{c} + \frac{\hat{n}\,\vec{r}\,'}{c}\right)$

$$\vec{A} = \frac{\mu_0}{4\pi r}\left\{\int d\vec{r}\,' \vec{j}\left(\vec{r}\,', \underbrace{t - \frac{r}{c}}_{t' \leftarrow \text{retarded time with respect to the origin}}\right) + \frac{\partial}{c\,\partial t'}\int d\vec{r}\,' (\hat{n}\,\vec{r}\,')\,\vec{j}(\vec{r}\,', t') + \dots\right\}$$

$$\vec{j}\left(\vec{r}\,', t - \frac{r}{c} + \frac{\hat{n}\,\vec{r}\,'}{c}\right) \approx j\left(\vec{r}\,', t - \frac{r}{c}\right) + \frac{\hat{n}\,\vec{r}\,'}{c}\frac{\partial}{\partial t'}\vec{j}(\vec{r}\,', t')$$

$$\int d\vec{r}\,' (\hat{n}\,\vec{r}\,') \sum q_j\,\vec{v}_j\,\delta(\vec{r}\,' - \vec{r}_j(t')) =$$

$$= \sum_j q_j\,\vec{v}_j\,(\hat{n}\,\vec{r}_j)$$

$$\vec{v}_j\,(\vec{r}_j\,\hat{n}) = \frac{1}{2}\frac{\partial}{\partial t'}\vec{r}_j\,(\hat{n}\,\vec{r}_j) + \underbrace{\frac{1}{2}\vec{v}_j\,(\hat{n}\,\vec{r}_j) - \frac{1}{2}\vec{r}_j\,(\hat{n}\,\vec{v}_j)}$$

$$\boxed{\frac{\partial}{\partial t}\vec{r}\,(\hat{n}\,\vec{r}) = \vec{v}\,(\hat{n}\,\vec{r}) + \vec{r}\,(\hat{n}\,\vec{v})}$$

$$\frac{1}{2}\;\hat{n}\times\vec{v}_j\times\vec{r}_j = \vec{v}_j\,(\hat{n}\cdot\vec{r}_j) - \vec{r}_j\,(\hat{n}\cdot\vec{v}_j)$$

$$= \frac{1}{2}\frac{\partial}{\partial t'}\vec{r}_j\,(\hat{n}\,\vec{r}_j) + \frac{1}{2}\underbrace{(\vec{r}_j\times\vec{v}_j)\times\hat{n}}$$

$$\Rightarrow \vec{A} = \frac{\mu_0}{4\pi r}\dot{\vec{d}} + \frac{1}{c}\frac{\partial}{\partial t'}\left\{\frac{1}{2}\frac{\partial}{\partial t'}\sum_j \vec{r}_j(\hat{n}\,\vec{r}_j)q_j\right.$$

$$\left. + \frac{1}{2}\sum_j q_j(\vec{r}_j \times \vec{v}_j)\times\hat{n}\right\}$$

j = particle index

From the definition of magnetic moment

$$\vec{m} = \frac{1}{2}\int \vec{r}\,' \times \vec{J}(\vec{r}\,')\,d^3\vec{r}\,' =$$

$$= \frac{1}{2}\sum_j q_j \int \vec{r}\,' \times \vec{v}_j\,\delta(\vec{r}\,' - \vec{r}_j)\,d^3\vec{r}\,' =$$

$$= \frac{1}{2}\sum_j q_j\,\vec{r}_j \times \vec{v}_j$$

$$\vec{A} = \frac{\mu_0}{4\pi r}\left\{\dot{\vec{d}} + \frac{1}{2c}\frac{\partial^2}{\partial t'^2}\sum_j\left[q_j\,\vec{r}_j(\hat{n}\cdot\vec{r}_j)\right]\right.$$

$$\left. + \frac{1}{c}\frac{\partial}{\partial t'}\,\vec{m}\times\hat{n}\right\}$$

can re-write

$$\frac{1}{3}\sum q_j\left(3\,\vec{r}_j(\hat{n}\cdot\vec{r}_j) - \hat{n}\,\vec{r}_j^{\,2}\right) = \frac{1}{3}\sum q\left(3x_\alpha x_\beta - \delta_{\alpha\beta}r^2\right)n_\beta$$

quadrupole moment Q

any term $\sim\hat{n}$ will disappear from $\vec{B} = \frac{1}{c}\dot{\vec{A}}\times\hat{n}$ and $\vec{E} = c\,\vec{B}\times\hat{n}$

α, β = vector indices

$$\cancel{\vec{A} = \frac{\mu_0}{4\pi r}\dot{\vec{d}} + \frac{1}{c}}$$

$$Q_{\alpha\beta} = \sum q\left(3x_\alpha x_\beta - \delta_{\alpha\beta}r^2\right) = \text{quadrupole moment}$$

form a vector $\vec{Q} = Q_\alpha\,\hat{e}_\alpha$, $Q_\alpha = Q_{\alpha\beta}\,n_\beta$

$$\vec{A} = \frac{\mu_0}{4\pi r}\left(\dot{\vec{d}} + \frac{1}{6c}\ddot{\vec{Q}} + \frac{1}{c}\dot{\vec{m}} \times \hat{n}\right)$$

$$\dot{\vec{A}} = \frac{\mu_0}{4\pi r}\left\{\ddot{\vec{d}} + \frac{1}{6c}\dddot{\vec{Q}} + \frac{1}{c}\ddot{\vec{m}} \times \hat{n}\right\}$$

Thus $$\vec{B} = \frac{1}{c}\dot{\vec{A}} \times \hat{n} = \underbrace{\frac{\mu_0}{4\pi c r}\ddot{\vec{d}} \times \hat{n}}_{\text{(electric dipole)}} + \underbrace{\frac{\mu_0}{4\pi c^2 r}\hat{n} \times (\hat{n} \times \ddot{\vec{m}})}_{\text{(magnetic dipole)}}$$

$$+ \underbrace{\frac{\mu_0}{24\pi c^2 r}\dddot{\vec{Q}} \times \hat{n}}_{\text{(electric quadrupole)}}$$

$$\vec{E} = c\,\vec{B} \times \hat{n}$$

All terms are $\frac{1}{r}$, but note additional $\frac{1}{c}$ in 2nd, 3rd.

Pointing vector $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = \frac{c}{\mu_0} \vec{B} \times \hat{n} \times \vec{B}$

$$\vec{E} \times \frac{1}{c} \hat{n} \times \vec{E} = \frac{\hat{n}}{c}\left(\vec{E}^2\right) - \frac{\vec{E}}{c}\left(\hat{n} \cdot \vec{E}\right) = \frac{1}{c} \hat{n} E^2$$

$$\vec{S} = \frac{1}{\mu_0 c} E^2 \hat{n} = \frac{1}{\mu_0 c} c^2 B^2 \hat{n}$$

$$= c\, u\, \hat{n} \quad , u = \frac{1}{2}\Big(\varepsilon_0 E^2 + \underbrace{\frac{1}{\mu_0} B^2}_{\frac{1}{\mu_0 c^2} E^2}\Big) = \varepsilon_0 E^2$$

$$\Rightarrow \vec{S} = c\, u\, \hat{n}.$$

also $\vec{S} = \frac{c}{\mu_0} \vec{B} \times \hat{n} \times \vec{B} = \frac{c}{\mu_0} \vec{B}^2$

$$\frac{1}{\mu_0 c} = \cancel{\mu_0} \frac{c}{\mu_0 c^2} = \varepsilon_0 c$$

Intensity into spherical angle $d\Omega$:

$$dI = \vec{S} \cdot d\vec{f} = S\, R_0^2\, d\Omega = c\, \varepsilon_0 E^2 R_0^2\, d\Omega$$

$$d\vec{f} = R_0^2\, d\Omega$$

Since $E, B \sim \frac{1}{R_0}$, $E^2 R_0^2 = \text{const}$

$\Rightarrow$ energy flows out with speed $\underline{c}$

$\Rightarrow$ radiation from accelerating charges.

Far-field ~~approx.~~ radiation and its angular distribution

$$\vec{A}(r,t) = \frac{\mu_0}{4\pi r}\left(\dot{\vec{d}} + \frac{1}{c}\,\dot{\vec{m}}\times\hat{n} + \frac{1}{6c}\,\ddot{\vec{Q}}\right)$$

$$\vec{B} = \frac{1}{c}\,\dot{\vec{A}}\times\hat{n} \quad , \quad \vec{E} = c\,\vec{B}\times\hat{n}$$

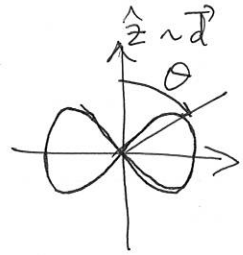
$$I = \frac{c}{\mu_0}\,\overline{\vec{B}^2} \quad \left(= \overline{\text{time-averaged intensity}}\right) = \frac{1}{T}\int_0^T dT \underbrace{\left(\frac{\vec{E}\times\vec{B}}{\mu_0}\right)}_{\vec{S}}$$

Radiation power emitted into solid angle $d\Omega$

$$\frac{dP}{d\Omega} = I\,R^2\,d\Omega$$

Total emitted power

$$P = \int \frac{dP}{d\Omega}\,d\Omega$$



profile of $\frac{dP}{d\Omega}$

Electric dipole

$$\vec{B}_{dipole} = \frac{\mu_0}{4\pi c r}\,\ddot{\vec{d}}\times\hat{n}$$

$$\vec{E}_{dip} = \frac{\mu_0}{4\pi c r}\left(\ddot{\vec{d}}\times\hat{n}\right)\times\hat{n} = \frac{\mu_0}{4\pi r}\left(\hat{n}\left(\ddot{\vec{d}}\cdot\hat{n}\right) - \ddot{\vec{d}}\right)$$

Charged particle $\vec{d} = q\vec{r} \Rightarrow \ddot{\vec{d}} = q\vec{a}$

$$\frac{dP}{d\Omega} = \frac{c}{\mu_0}\left(\frac{\mu_0}{4\pi c}\right)^2 \overline{\left(\ddot{\vec{d}}\times\hat{n}\right)^2} = \frac{\mu_0}{16\pi^2 c}\,\overline{\left(\ddot{\vec{d}}\times\hat{n}\right)^2} \sim \sin^2\theta$$

$$\left(\ddot{\vec{d}}\times\hat{n}\right)^2 = \left(\ddot{\vec{d}}\times\hat{n}\right)\left(\ddot{\vec{d}}\times\hat{n}\right) = \ddot{\vec{d}}\cdot\left[\hat{n}\times\left(\ddot{\vec{d}}\times\hat{n}\right)\right] = \ddot{\vec{d}}\cdot\left[\ddot{\vec{d}} - \hat{n}\left(\hat{n}\,\ddot{\vec{d}}\right)\right]$$

$$A\cdot B\times C = B\cdot C\times A = C\cdot A\times B$$

$$\begin{vmatrix} A \\ B \\ C \end{vmatrix} = -\begin{vmatrix} B \\ A \\ C \end{vmatrix} = \begin{vmatrix} B \\ C \\ A \end{vmatrix}$$

$$\left(\ddot{\vec{d}}\times\hat{n}\right)^2 = \ddot{\vec{d}}^{\,2} - \left(\hat{n}\cdot\ddot{\vec{d}}\right)^2 = \ddot{\vec{d}}^{\,2}\left(1-\cos^2\theta\right) = \ddot{\vec{d}}^{\,2}\sin^2\theta$$

choosing $\ddot{\vec{d}}$ along $\hat{z}$ axis

Magnetic dipole

$$\vec{B} = \frac{1}{c^2}\left(\ddot{\vec{m}} \times \hat{n}\right) \times \hat{n} \left(\frac{\mu_0}{4\pi r}\right) = \frac{-\mu_0}{4\pi r c^2} \left(\hat{n} \times \ddot{\vec{m}}\right) \times \hat{n}$$

$$= -\frac{\mu_0}{4\pi r c^2}\left(\ddot{\vec{m}} - \hat{n}\left(\hat{n} \cdot \ddot{\vec{m}}\right)\right)$$

$$\left.\begin{aligned} \vec{B}_{md} &= \frac{\mu_0}{4\pi r c^2}\left(\hat{n}\left(\hat{n} \cdot \ddot{\vec{m}}\right) - \ddot{\vec{m}}\right) \\ \vec{E}_{md} &= c\,\vec{B} \times \hat{n} = \frac{\mu_0}{4\pi r c}\left(-\ddot{\vec{m}} \times \hat{n}\right) \end{aligned}\right\}$$

very similar in form to electric dipole fields

Namely $\vec{d} \to \frac{\vec{m}}{c}$, $\vec{E} \to c\vec{B}$, $\vec{B} \to -\frac{\vec{E}}{c}$

$$\begin{cases} \vec{E}_{dip} = \frac{\mu_0}{4\pi r}\left(\hat{n}\left(\ddot{\vec{d}} \cdot \hat{n}\right) - \ddot{\vec{d}}\right) \longrightarrow c\,\vec{B}_{md} = \frac{\mu_0}{4\pi r c}\left(\hat{n}\left(\ddot{\vec{m}} \cdot \hat{n}\right) - \ddot{\vec{m}}\right) \\ \vec{B}_{dip} = \frac{\mu_0}{4\pi c r}\,\ddot{\vec{d}} \times \hat{n} \longrightarrow -\frac{\vec{E}_{md}}{c} = \frac{\mu_0}{4\pi c^2 r}\,\ddot{\vec{m}} \times \hat{n} \end{cases}$$

$$\vec{E}_{md} = +\frac{\mu_0}{4\pi c r}\left(\hat{n} \times \ddot{\vec{m}}\right)$$

$$I = \frac{c}{\mu_0}\overline{\vec{B}_{dip}^{\,2}} \longrightarrow \frac{c}{\mu_0}\overline{\left(\frac{\vec{E}_{md}}{c}\right)^2} = \frac{c}{\mu_0 c}\left(\frac{\mu_0}{4\pi c r}\right)^2 \overline{\left(\hat{n} \times \ddot{\vec{m}}\right)^2}$$

$$\frac{dP_{md}}{d\Omega} = \frac{\mu_0}{16\pi^2 c^3}\overline{\left(\ddot{\vec{m}} \times \hat{n}\right)^2}$$

Electric quadrupole

$$\vec{A} = \frac{\mu_0}{24\pi c r} \ddot{\vec{Q}} \quad , \quad \ddot{\vec{Q}} = \ddot{\overleftrightarrow{Q}}\,\hat{n}$$

i.e. $Q_\alpha = Q_{\alpha\beta}\, n_\beta$

$$\vec{B} = \frac{\mu_0}{24\pi c^2 r} \dddot{\vec{Q}} \times \hat{n}$$

$$\frac{dP}{d\Omega} = \frac{c}{\mu_0}\left(\frac{\mu_0}{24\pi c^2}\right)^2 \overline{\left(\dddot{\vec{Q}} \times \hat{n}\right)^2} = \frac{\mu_0}{24^2\pi^2 c^3}\overline{\left[\dddot{\vec{Q}}^2 - \left(\hat{n}\cdot\dddot{\vec{Q}}\right)^2\right]}$$

$$\dddot{\vec{Q}}^2 = \sum_\alpha \dddot{Q}_\alpha^{\,2} = \sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\, n_\beta\, \dddot{Q}_{\alpha\gamma}\, n_\gamma$$

$$\hat{n}\cdot\dddot{\vec{Q}} = \sum_\alpha \dddot{Q}_\alpha\, n_\alpha = \sum_{\alpha\beta} \dddot{Q}_{\alpha\beta}\, n_\alpha n_\beta$$

$$\frac{dP}{d\Omega} = \frac{\mu_0}{576\pi^2 c^3}\overline{\left\{\sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\gamma}\, n_\beta n_\gamma - \sum_{\alpha\beta\gamma\delta} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\gamma\delta}\, n_\alpha n_\beta n_\gamma n_\delta\right\}}$$

Total emitted power

$$P = \int \frac{dP}{d\Omega}\, d\Omega \Rightarrow \text{we need to average}$$

$$\underbrace{\frac{1}{4\pi}\int d\Omega\, n_\beta n_\gamma}_{\frac{1}{3}} \quad \text{and} \quad \underbrace{\int \frac{d\Omega}{4\pi}\, n_\alpha n_\beta n_\gamma n_\delta}_{\frac{1}{15}}$$

(see next page)